Advanced Machine Learning: Stage A Results. Overparameterization and Generalization in Speech Recognition

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1 Introduction

Classical statistical learning theory predicts that model capacity beyond the interpolation threshold should cause generalization to degrade. Modern overparameterized models frequently improve after that threshold, a phenomenon described by the double-descent risk curve [1, 2]. The effect is documented mainly in image classification [2]; its behavior in sequence-to-sequence tasks such as automatic speech recognition (ASR) is less studied.

Stage A covers three experiments for a project investigating double descent in a Conformer-CTC [3, 4] ASR system trained on LibriSpeech: a preliminary study of input normalization, a capacity sweep on the full training set, and a subset ablation that reduces the corpus to place the interpolation threshold within the evaluated capacity range.

Throughout, the interpolation threshold is $\mathcal{L}_{\text{train}} < 0.01$, following the overparameterization theory of Belkin et al. [1].

2 Experimental Setup

2.1 Model

All experiments use a Conformer-CTC architecture [4]. Each Conformer block contains a feed-forward module, multi-head self-attention, a convolution module with BatchNorm, and a second feed-forward module, connected with pre-norm residual connections and five layer-normalization operations per block. The CTC head is a single linear projection onto the vocabulary. The capacity dimension varied in Stage A is the number of Conformer blocks $L \in \{1, 2, 3, 4, 6, 8, 12, 16\}$, with hidden dimension $d = 512$ and $h = 8$ attention heads fixed. Parameter counts range from 14 M at $L = 1$ to 105 M at $L = 16$.

2.2 Data

Models are trained on LibriSpeech `train-clean-360` and `train-clean-100` combined (≈460 h, 460 k utterances). The `dev-clean` split serves as the validation set and is evaluated at every checkpoint during training; it is not used for any hyperparameter selection in Stage A. `test-clean` is held out entirely during training and used only for final evaluation of each completed run. Audio is represented as 80-dimensional log-mel spectrograms ($n_{\text{fft}} = 400$, hop=160, window=400 at 16 kHz). Targets are SentencePiece unigram tokens with a vocabulary of 1 000 pieces. Utterance-level CMVN is applied by default unless stated otherwise.

For the subset ablation study (Section 5), training is restricted to the first $N \in \{5\text{k}, 50\text{k}, 150\text{k}\}$ utterances drawn from the combined corpus, with all other settings identical to the full-data runs.

2.3 Training

All runs use the Adam optimiser with a warm-up cosine learning rate schedule ($\text{LR}_{\text{max}} = 10^{-4}$, $\text{LR}_{\text{min}} = 10^{-6}$, 4 k warm-up steps), weight decay 10^{-6}, batch size 16, and dropout disabled. Training runs for 100 000 optimiser steps. Models are evaluated every 5 000 steps on a fixed subset of up to 500 batches drawn from each split. WER is computed by greedy CTC decoding.

2.4 Logged Metrics

Sharpness proxy. A one-sided random-perturbation estimate of loss curvature:

$$\widehat{S} \;=\; \mathbb{E}_{\delta}[\max(0,\, \mathcal{L}(\theta+\delta) - \mathcal{L}(\theta))]\,, \qquad \delta = \frac{\varepsilon_{\text{rel}}\,\|\theta\|_2}{\|\xi\|_2}\,\xi, \quad \xi \sim \mathcal{N}(0, I), \tag{1}$$

where $\varepsilon_{\text{rel}} = 10^{-3}$. The expectation is approximated over $N_b = 100$ batches and $N_p = 3$ independent perturbations per batch. This metric is logged as an auxiliary indicator only. The one-sided form retains only perturbations that increase the loss, avoiding contamination from flat or downhill directions. The dense normalization stack (5×LayerNorm + BatchNorm per block) attenuates the effect of weight perturbations at the output, making the proxy unreliable as an absolute curvature estimate for this architecture; a more robust measure (filter-normalized or SAM-based) will be considered for Stage B.

Relative sharpness.

$$\widehat{S}_{\text{rel}} \;=\; \frac{\widehat{S}}{\mathcal{L}_{\text{test}} + \epsilon}, \qquad \epsilon = 10^{-12}. \tag{2}$$

Weight norm.

$$\|\theta\|_W \;=\; \sqrt{\sum_{p\in\theta} \|p\|_2^2}. \tag{3}$$

Generalization gap.

$$\Delta \;=\; \mathcal{L}_{\text{test}} - \mathcal{L}_{\text{train-eval}}. \tag{4}$$

$\mathcal{L}_{\text{train-eval}}$ is evaluated on a fixed random subset of the training set at each checkpoint, decoupled from the running training loss.

3 Preliminary study: effect of CMVN mode

Before the main capacity sweep we verified whether input normalization affects training dynamics or final performance, as differences here could confound the interpretation of Stage A results. Three CMVN modes were compared on a fixed $d = 512$, $L = 8$ (57 M parameter) model: none (raw log-mel), global (per-bin training-set statistics), and utt (per-utterance mean–variance normalization over time frames). Each condition was run with two random seeds. Performance is evaluated on validation split.

Table 1: CMVN comparison at 100 k steps (mean over 2 seeds, evaluated on `dev-clean`).

Mode	Val WER (%)	Best Val WER (%)	Gen. Gap
none	11.48	**11.45**	0.593
global	11.61	11.53	0.573
utt	11.66	11.60	0.599

All three modes yield virtually identical training dynamics and final performance (Table 1). The largest difference in val WER across conditions is 0.18 percentage points, well within the inter-seed variance. Loss and WER curves for both splits are visually indistinguishable across conditions (Figure 1, 2).

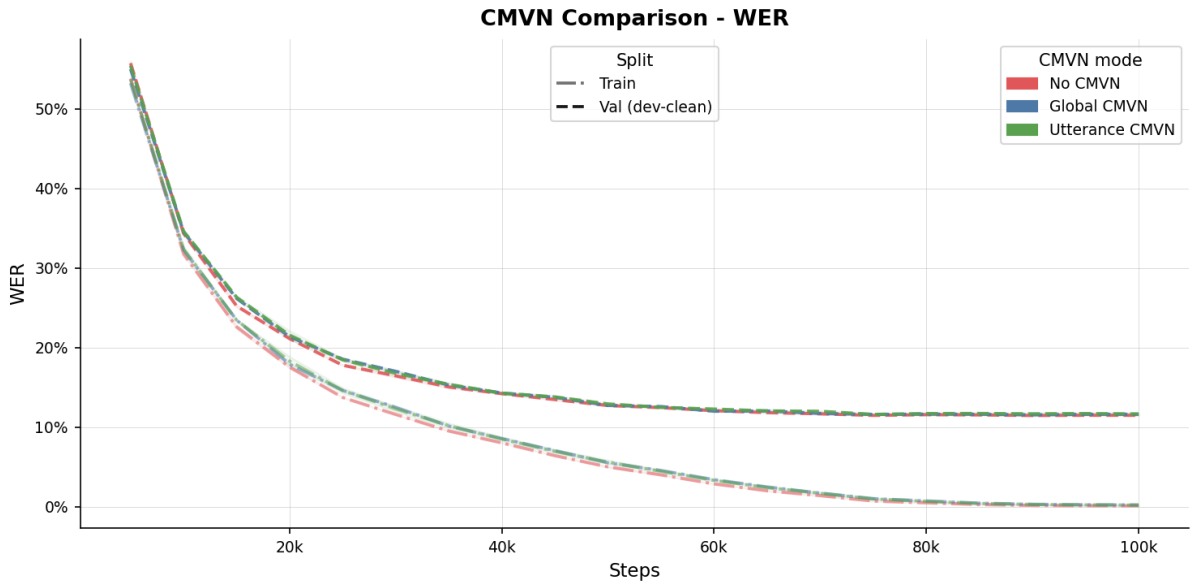

Figure 1: WER over training steps by CMVN mode.

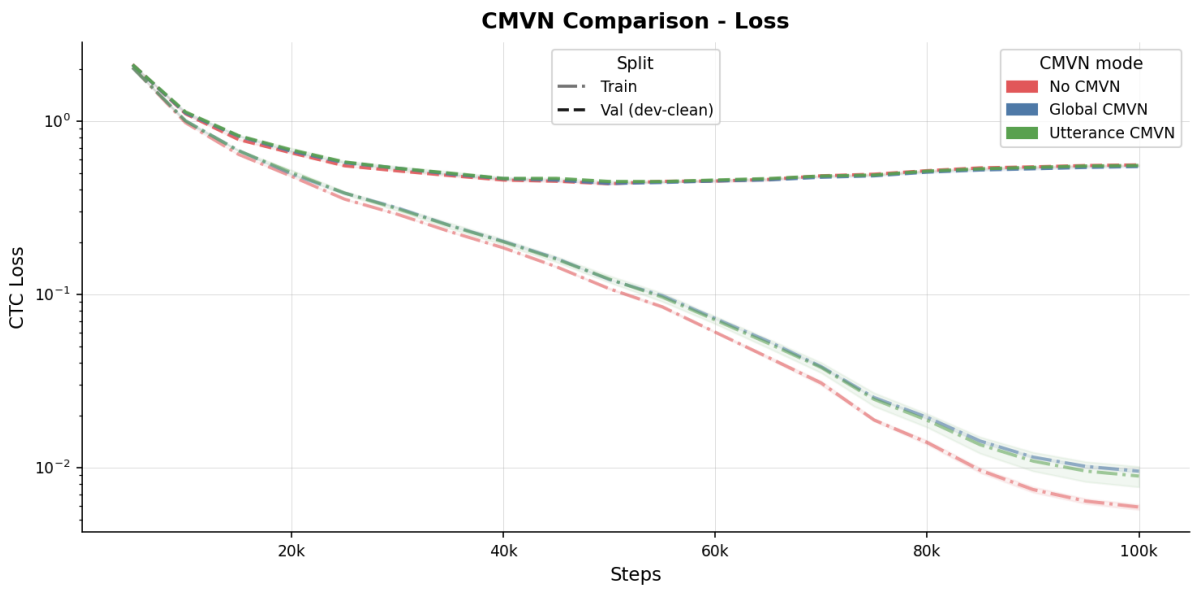

Figure 2: CTC loss over training steps by CMVN mode (log scale).

The dense internal normalization stack (LayerNorm, BatchNorm) likely absorbs input-distribution shifts, which would explain the result, though other confounders cannot be excluded without ablating those components. Regardless of the mechanism, the result justifies any consistent CMVN mode across runs; all Stage A experiments use utt.

4 Stage A: capacity sweep

The central hypothesis of this project is that overparameterized ASR models exhibit a double-descent generalization curve: as model capacity increases past the interpolation threshold, test error first peaks and then decreases. Stage A establishes the baseline capacity curve under clean training labels. The depth L is varied while $d = 512$ is held fixed, eliminating the confound between width and depth present in earlier pilot runs. Two random seeds are used per configuration; all training settings follow Section 2.3.

At step 100 k, models with $L \geq 8$ satisfy $\mathcal{L}_{\text{train}} \leq 0.009$ and thus meet the interpolation criterion; smaller models retain non-trivial training loss.

4.1 Results

Capacity vs. generalization. Test WER decreases monotonically with model size across all evaluated configurations (Table 2, Figures 3–4). Test loss follows the same overall trend but has a slight non-monotonicity at $L = 4$ (0.692 vs. 0.680 for $L = 3$) before resuming its decrease. No double-descent peak is visible in WER under clean labels within the 100 k step training budget. The generalization gap (Table 2) is highest for the non-interpolating models ($L \leq 2$) and shows a local peak at $L = 4$ (0.660) relative to $L = 3$ (0.592) before decreasing for larger models, a pattern consistent with the onset of the overparameterization regime.

Table 2: Stage A results on the full training set (mean over 2 seeds). All loss and WER values are at step 100 k.

L	Params	Train WER	Test WER	$\mathcal{L}_{\text{train}}$	$\mathcal{L}_{\text{test}}$	Gen. Gap	$\|\theta\|_W$
1	14 M	19.1%	39.0%	0.423	1.258	0.836	137
2	20 M	7.6%	23.0%	0.173	0.798	0.626	171
3	26 M	3.7%	17.7%	0.088	0.680	0.592	198
4	32 M	1.1%	15.9%	0.032	0.692	0.660	222
6	44 M	0.5%	13.3%	0.018	0.626	0.608	260
8	57 M	0.2%	12.1%	0.009	0.607	0.597	293
12	81 M	0.1%	11.0%	0.004	0.580	0.576	346
16	105 M	0.0%	10.3%	0.002	0.571	0.568	390

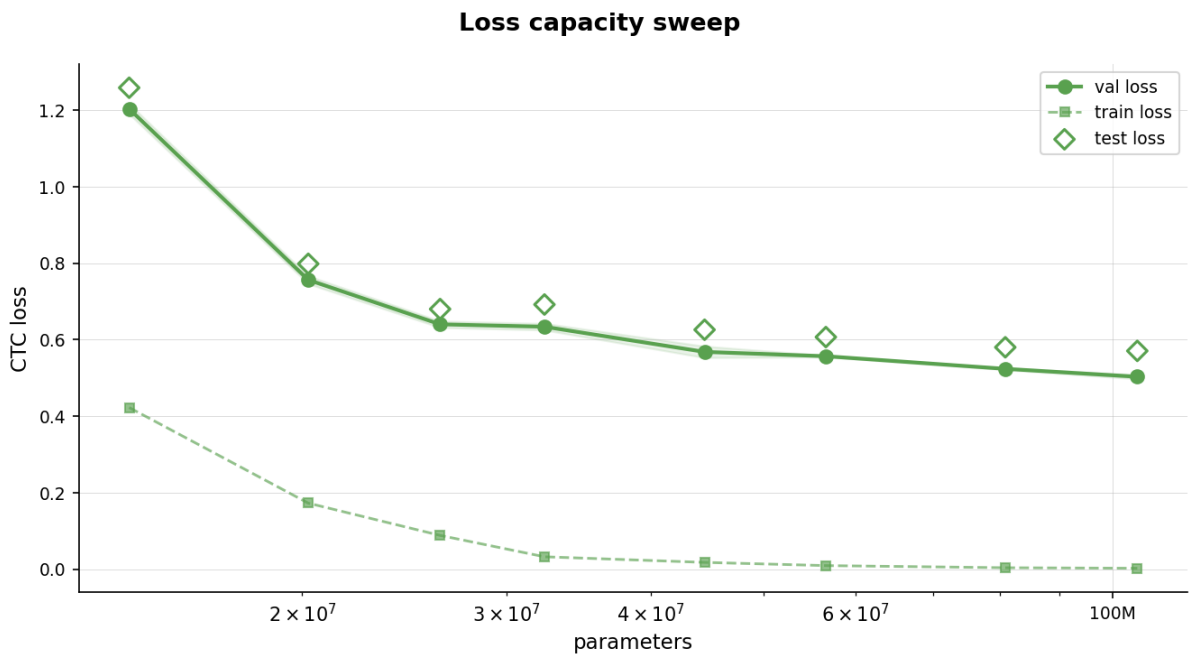

Figure 3: Train-eval loss and test loss at step 100 k as a function of model capacity (full training set). Train-eval loss decreases monotonically; test loss and val loss show a slight non-monotonicity at $L = 4$ before resuming their decrease. Shaded band: seed range.

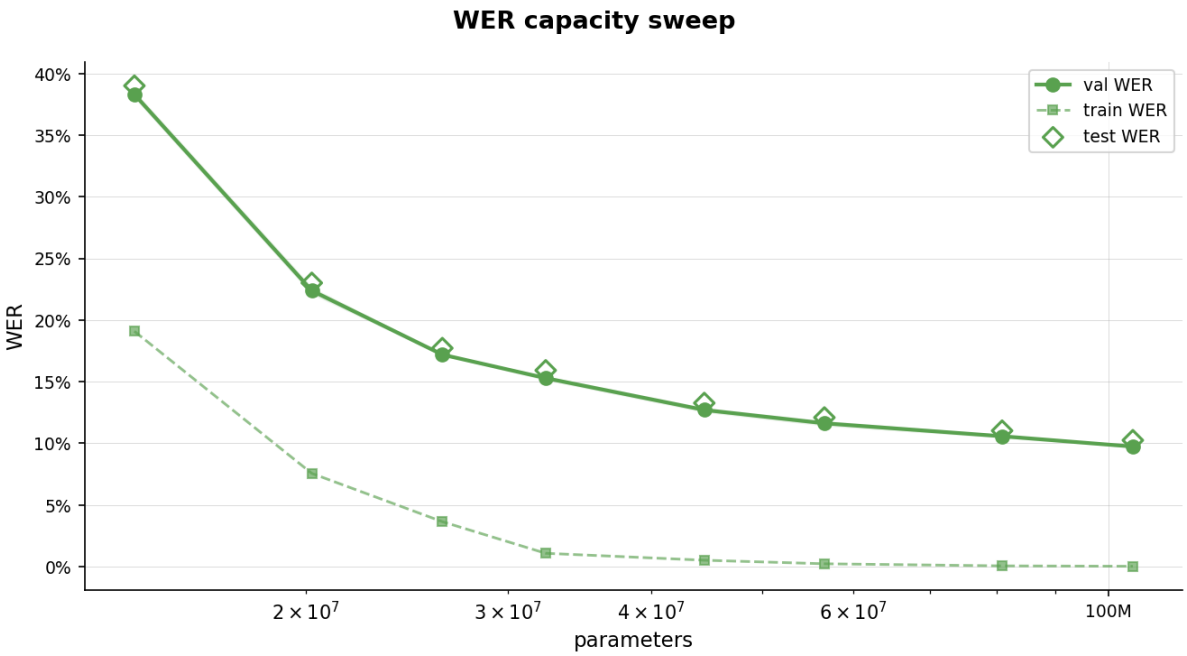

Figure 4: Train WER, val WER (dev-clean), and test WER (diamonds) vs. parameter count at step 100 k (full training set). Shaded band: seed range.

Epoch-wise trajectory. Val loss rises once models begin to interpolate (Figures 5–6), consistent with classical overfitting. Val WER diverges from this trend: larger models keep reducing WER long after training loss reaches near zero. Shallower models ($L = 4$, $L = 8$) hit a val WER minimum at around step 75 k and rise slightly; deeper models ($L \geq 12$) keep improving through step 100 k. Greedy CTC decoding is a threshold operation, so argmax improvements can coexist with rising cross-entropy.

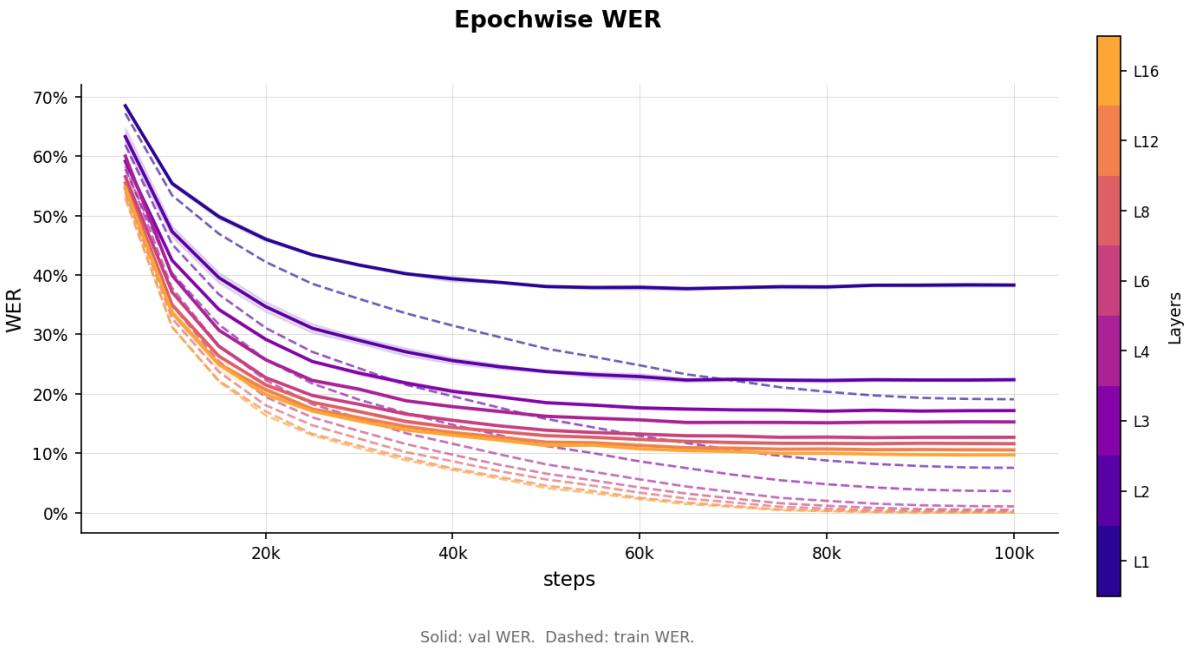

Figure 5: Val WER (solid) and train WER (dashed) over training steps for all capacity levels (full training set). Shaded band: seed range.

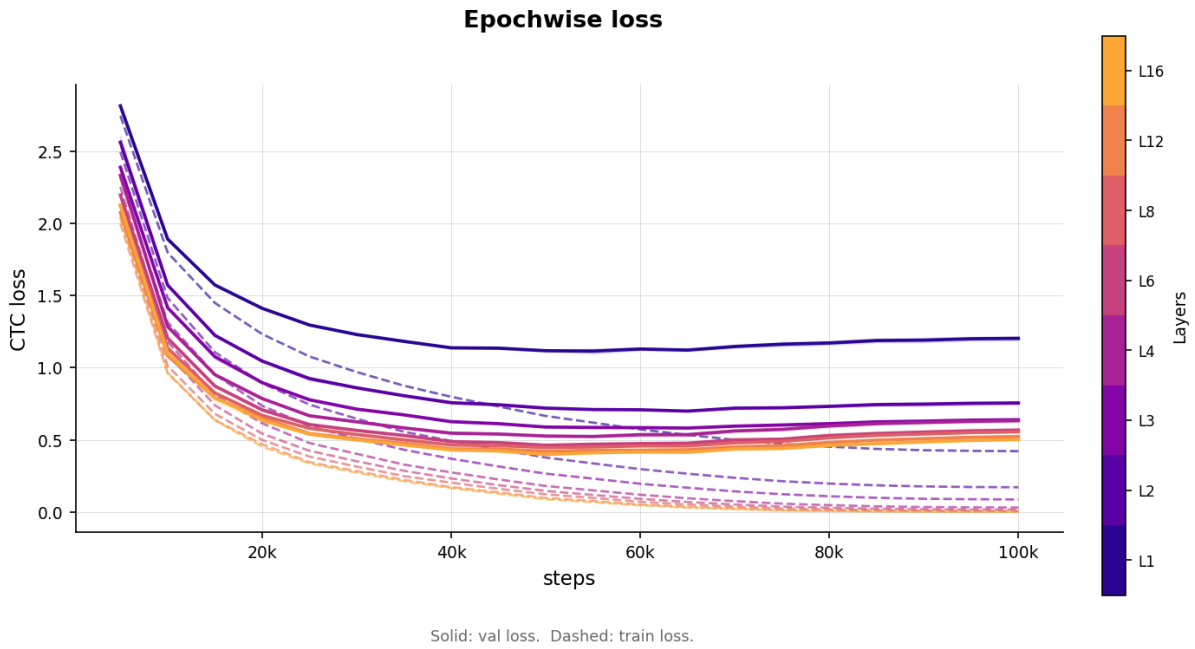

Figure 6: Val loss (solid) and train-eval loss (dashed) over training steps. Val loss rises after interpolation while train loss approaches zero. Shaded band: seed range.

Sharpness and weight norm. Final relative sharpness values ($\widehat{S}_{\text{rel}}$) are uniformly small ($\approx 10^{-4}$) across all model sizes with no consistent ordering by depth (see Section 2.4 for a discussion of the proxy's limitations). The metric is retained as a baseline for Stage B comparisons.

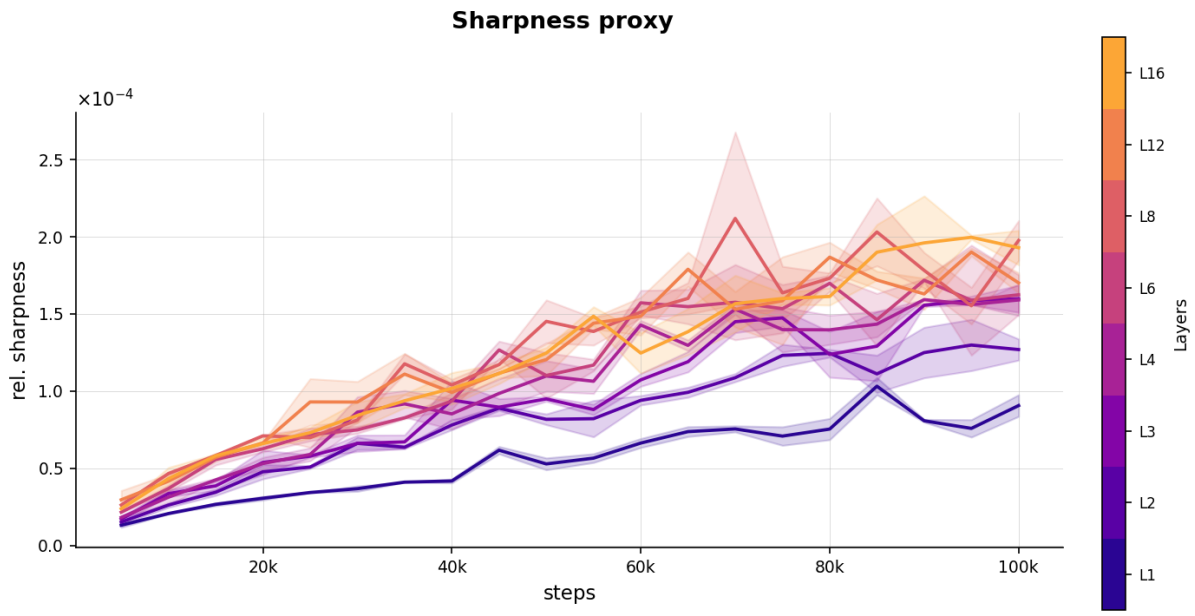

Figure 7: Relative sharpness proxy ($\widehat{S}_{\text{rel}}$, Eq. 2) over training steps (full training set). Shaded band: seed range.

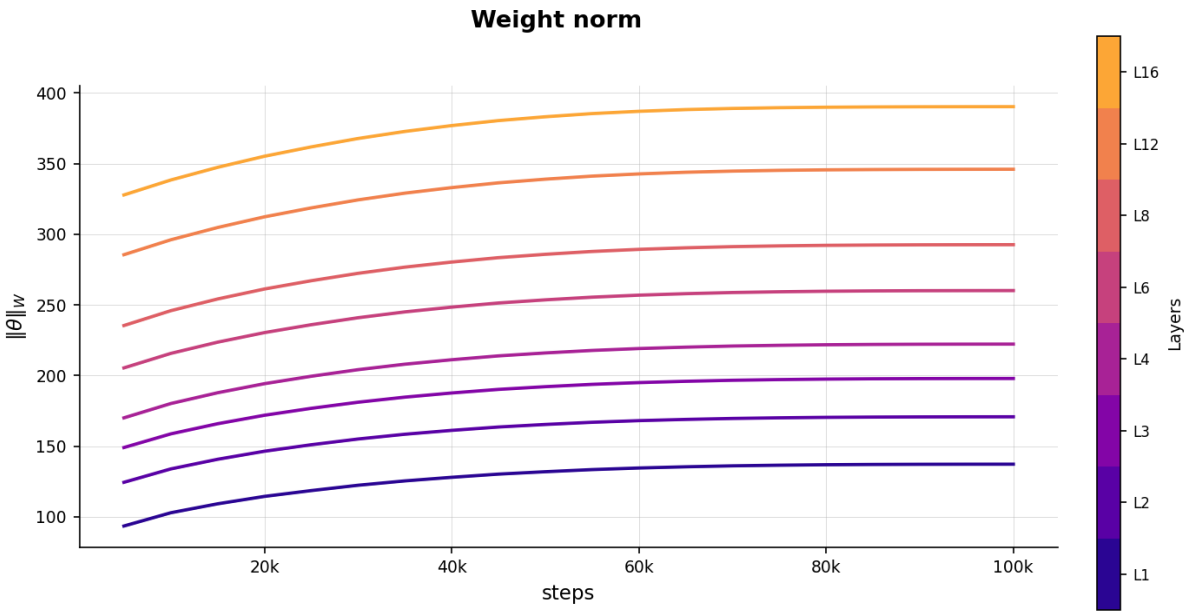

Figure 8: Weight norm $\|\theta\|_W$ over training steps (full training set). Shaded band: seed range.

Reproducibility. Inter-seed variation in final test WER is below 0.45 percentage points for all configurations.

4.2 Discussion

Double descent requires a training set near the interpolation threshold, or label noise [1]. On 460 h of clean speech, $L = 16$ reaches near-interpolation ($\mathcal{L}_{\text{train}} = 0.002$) but the dataset is too large for a generalization peak to appear.

Val WER keeps improving long after training loss approaches zero (Figures 5–6). For $L \geq 12$ the curve has not turned by step 100 k; a longer training budget would likely reveal the full benefit of overparameterization.

5 Stage A: subset size ablation study

The capacity sweep was repeated with training corpora restricted to $N \in \{5\text{k}, 50\text{k}, 150\text{k}\}$ utterances, to probe two complementary regimes: one where every model overfits the training corpus, and one where smaller models underfit. Together these locate the experimental setup relative to the interpolation threshold across a range of data scales.

5.1 Small subset results

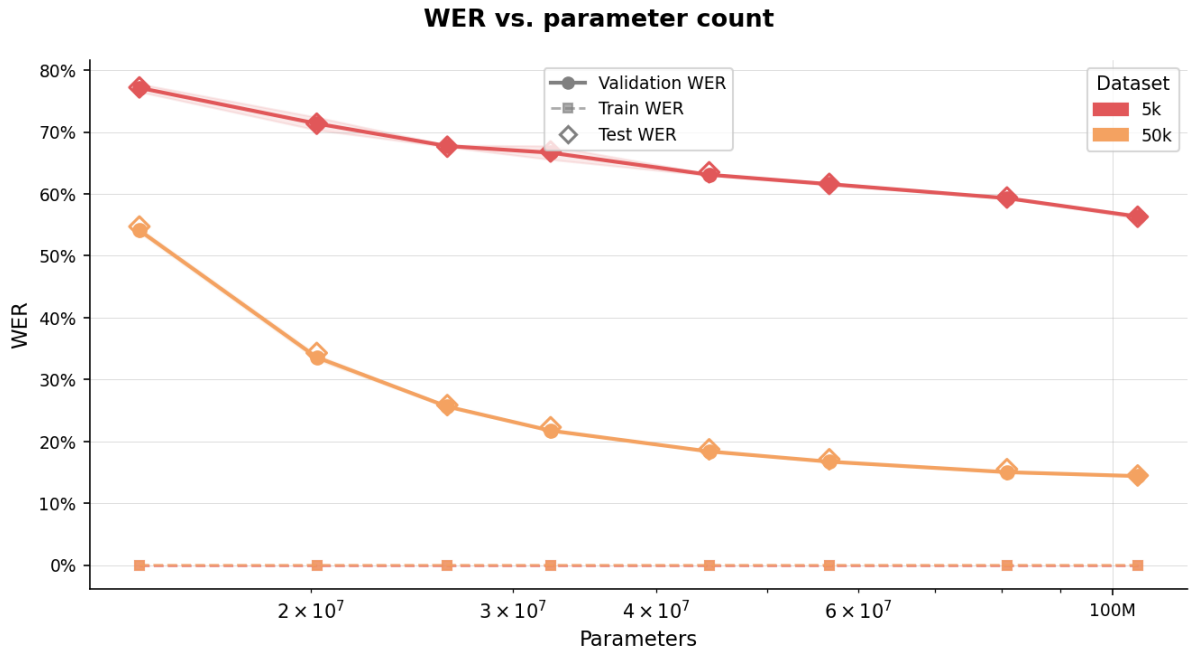

Figure 9: WER vs. parameter count at step 100 k for the small subsets (5k and 50k utterances). All models reach train WER = 0%; larger models nonetheless achieve lower val and test WER.

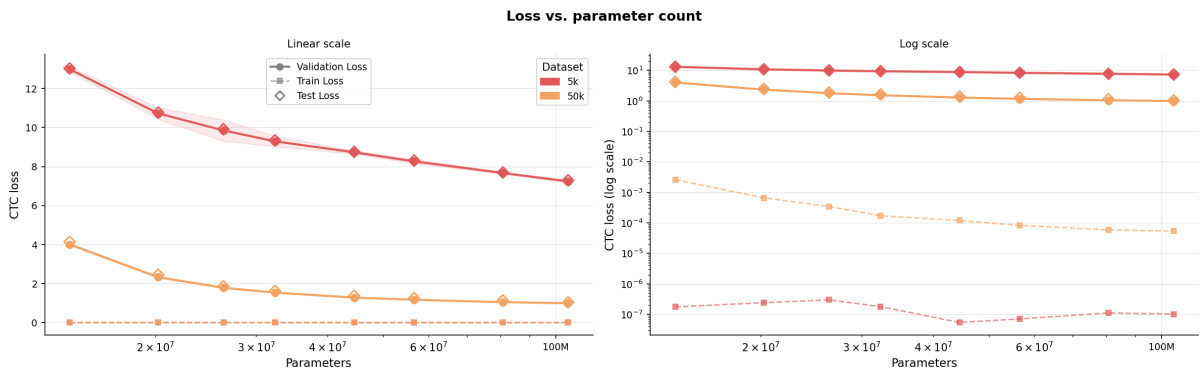

Figure 10: Loss vs. parameter count at step 100 k for the small subsets. Train-eval loss is near zero for all models. Shaded band: seed range.

With 5k and 50k training utterances, every model reaches near-zero training loss and training WER before the end of training (Table 3), including $L = 1$ (14 M parameters). All models are firmly in the overparameterization regime. Yet val WER decreases monotonically from $L = 1$ to $L = 16$: larger models generalize better despite memorizing the same training set.

Table 3: Final step (100 k) metrics for the small subsets. All models reach train WER = 0% and $\mathcal{L}_{\text{train}} \approx 0$.

		5k utterances			50k utterances		
L	Params	Val WER	Test WER	Gen. Gap	Val WER	Test WER	Gen. Gap
1	14 M	77.2%	77.2%	13.03	54.1%	54.7%	4.14
2	20 M	71.4%	71.4%	10.77	33.6%	34.3%	2.44
3	26 M	67.7%	67.6%	9.91	25.6%	25.9%	1.85
4	32 M	66.7%	66.9%	9.34	21.7%	22.3%	1.61
6	44 M	63.1%	63.5%	8.75	18.4%	18.7%	1.37
8	57 M	61.6%	61.6%	8.30	16.7%	17.1%	1.25
12	81 M	59.3%	59.4%	7.69	15.0%	15.5%	1.13
16	105 M	56.4%	56.3%	7.29	14.4%	14.5%	1.03

No double-descent peak appears: $L = 1$ (14 M parameters) fully memorizes 5k or 50k utterances, so no model underfits these corpora. Only the descending branch is accessible; seeing the peak requires either much smaller models or an even smaller training corpus.

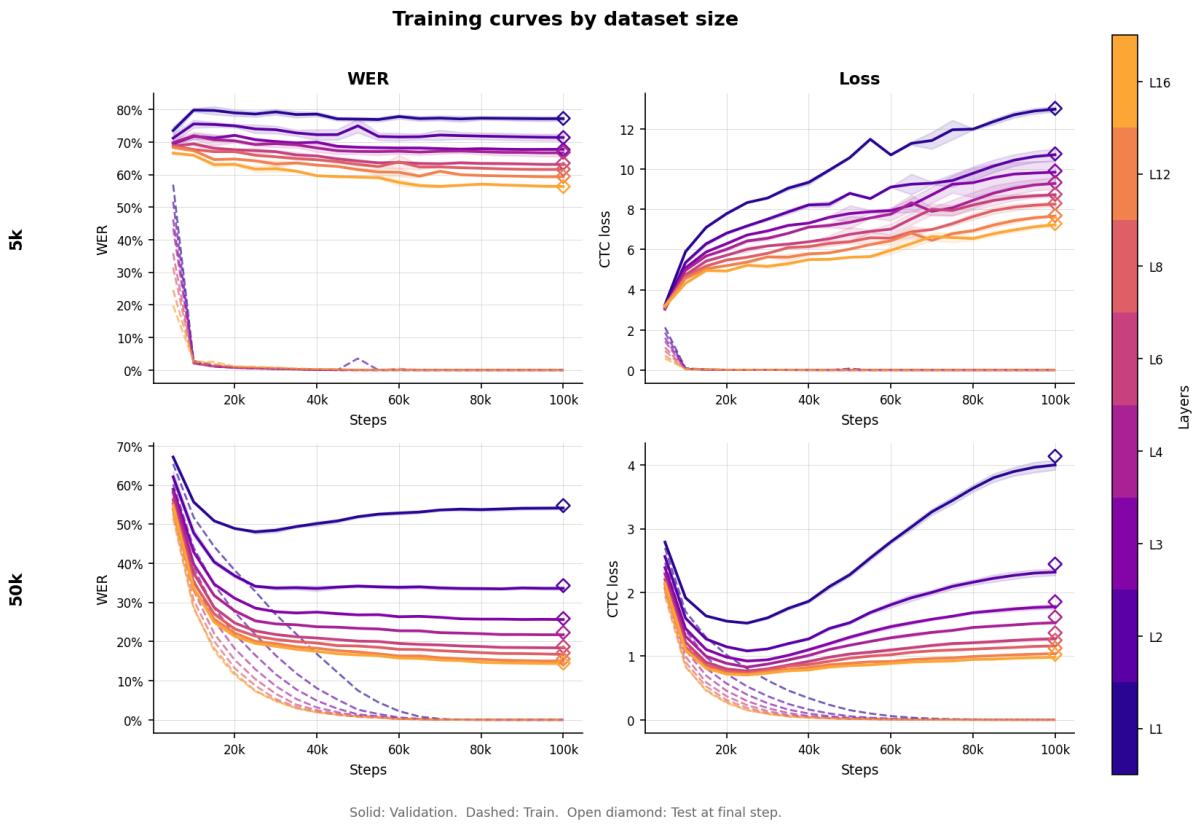

Figure 11: Training curves (WER, left; loss, right) for the small subsets. Each row is one subset size; color indicates model depth. Solid: validation. Dashed: train. Open diamond: test WER at final step.

5.2 Large subset results

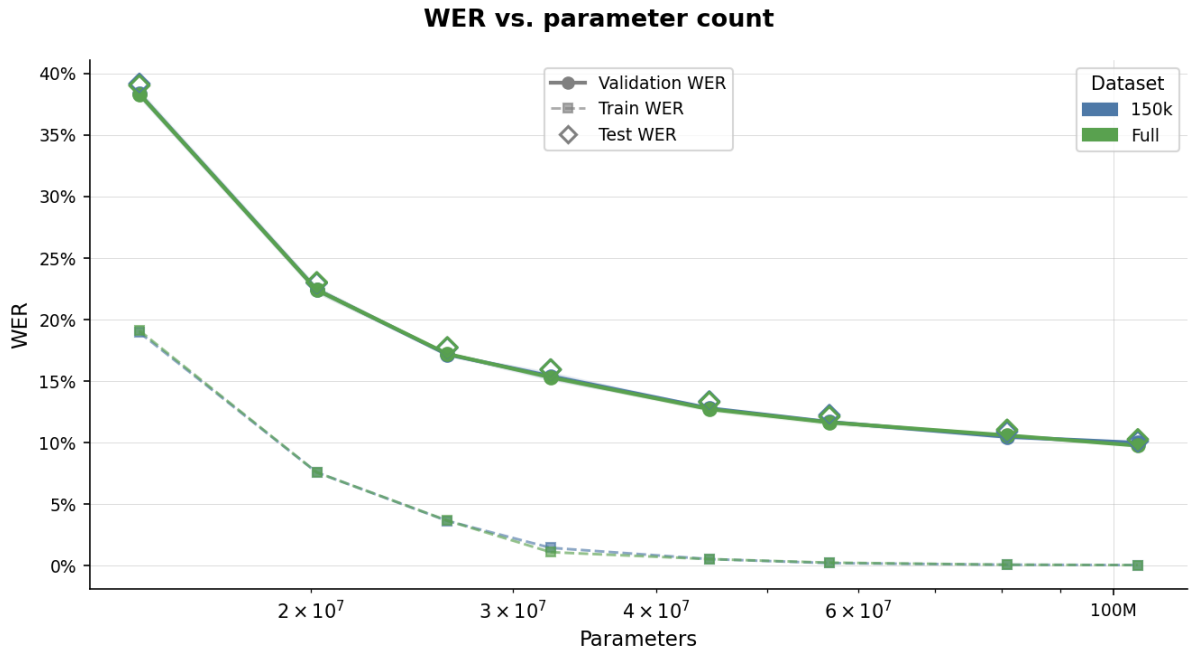

Figure 12: WER vs. parameter count at step 100 k for the large subsets (150k utterances and full dataset). Small models ($L \leq 2$) remain in the underfitting regime; larger models approach near-zero training loss. Shaded band: seed range.

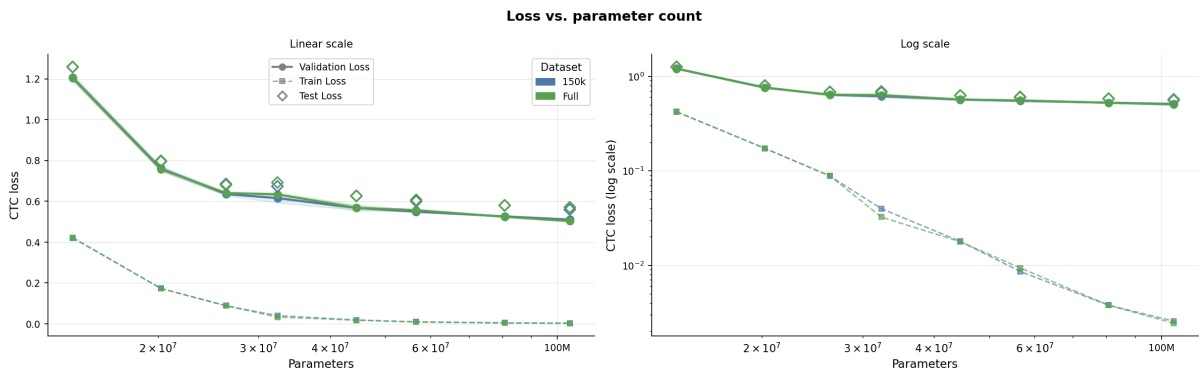

Figure 13: Loss vs. parameter count at step 100 k for the large subsets. Train-eval loss decreases with depth but remains non-trivial for small models. Shaded band: seed range.

With 150k or the full ∼460k utterances, small models ($L = 1$, $L = 2$) fail to fit the training data within 100 k steps (train WER 19%, 8%), while larger models progressively approach near-zero training loss. The 150k results are closely consistent with the full-dataset sweep; both are summarized in Table 4. The generalization gap (Table 4) mirrors the pattern from Section 4.1: it peaks at $L = 4$ for the 150k subset (0.634 vs. 0.597 at $L = 3$) before decreasing monotonically for larger models, consistent with the onset of the overparameterization regime.

Table 4: Final step (100 k) metrics for the large subsets.

		150k utterances				Full (≈460k)			
L	Params	Train WER	Test WER	$\mathcal{L}_{\text{train}}$	Gen. Gap	Train WER	Test WER	$\mathcal{L}_{\text{train}}$	Gen. Gap
1	14 M	18.9%	39.1%	0.420	0.837	19.1%	39.0%	0.423	0.836
2	20 M	7.6%	23.0%	0.173	0.623	7.6%	23.0%	0.173	0.626
3	26 M	3.6%	17.7%	0.088	0.597	3.7%	17.7%	0.088	0.592
4	32 M	1.4%	15.9%	0.040	0.634	1.1%	15.9%	0.032	0.660
6	44 M	0.5%	13.3%	0.018	0.609	0.5%	13.3%	0.018	0.608
8	57 M	0.2%	12.2%	0.009	0.592	0.2%	12.1%	0.009	0.597
12	81 M	0.1%	10.9%	0.004	0.576	0.1%	11.0%	0.004	0.576
16	105 M	0.0%	10.1%	0.003	0.557	0.0%	10.3%	0.002	0.568

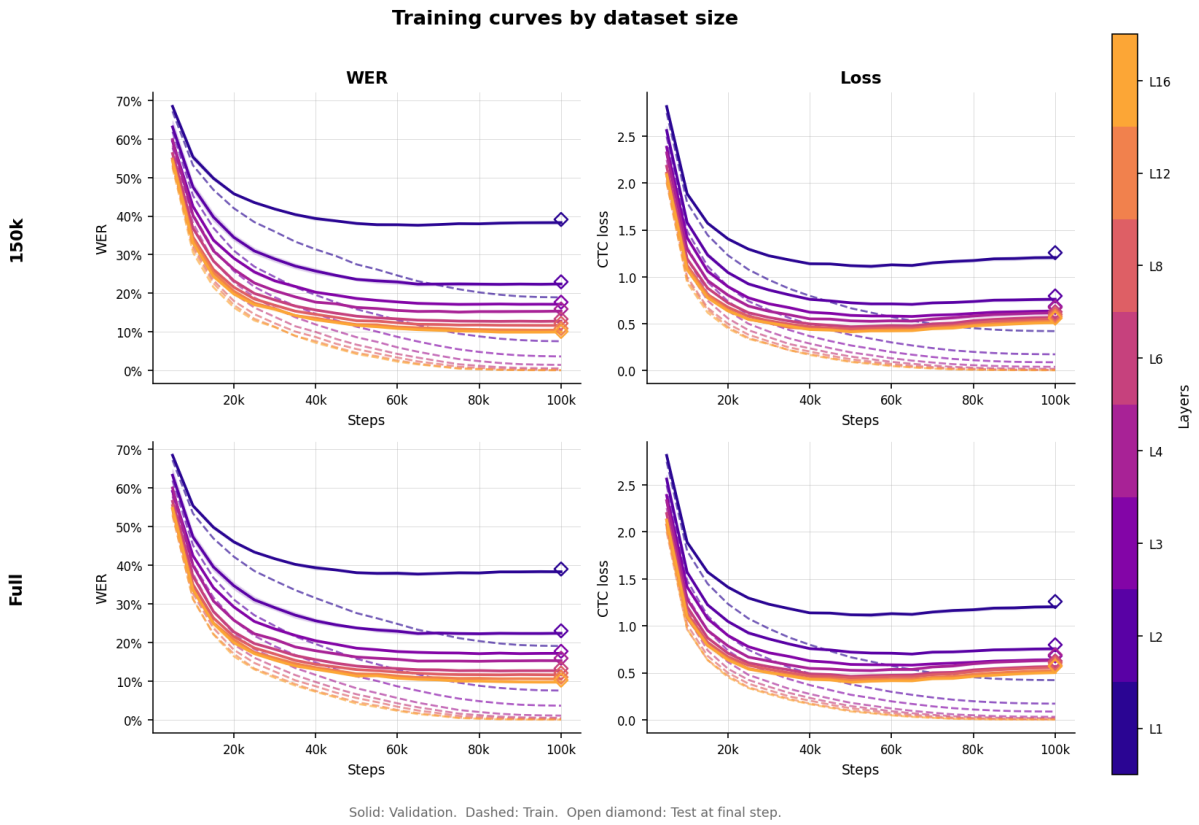

Figure 14: Training curves (WER, left; loss, right) for the large subsets. Each row is one subset size; color indicates model depth. Solid: validation. Dashed: train. Open diamond: test WER at final step.

These runs sit on the underfitting side of the capacity curve. No double-descent peak appears here either: $L = 16$ reaches near-interpolation ($\mathcal{L}_{\text{train}} = 0.002$), but we observe no generalization penalty within 100 k steps. The model family spans the transition from underfitting to near-interpolation but does not clearly enter the overparameterization regime within the current training budget.

5.3 Cross-subset overview

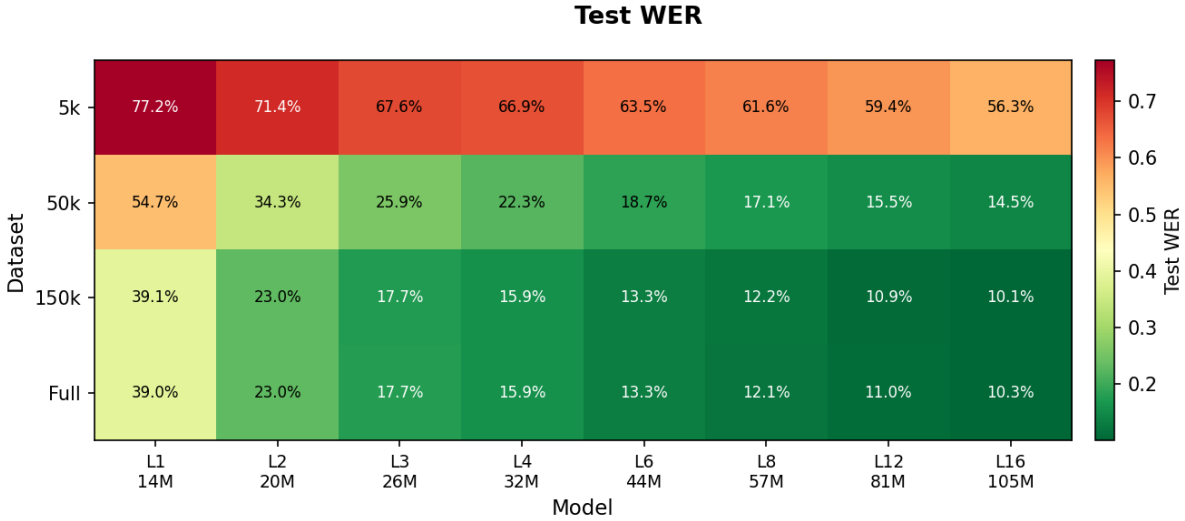

Figure 15: Test WER at step 100 k across all subset sizes (rows) and model capacities (columns). Larger models consistently achieve lower test WER within each row.

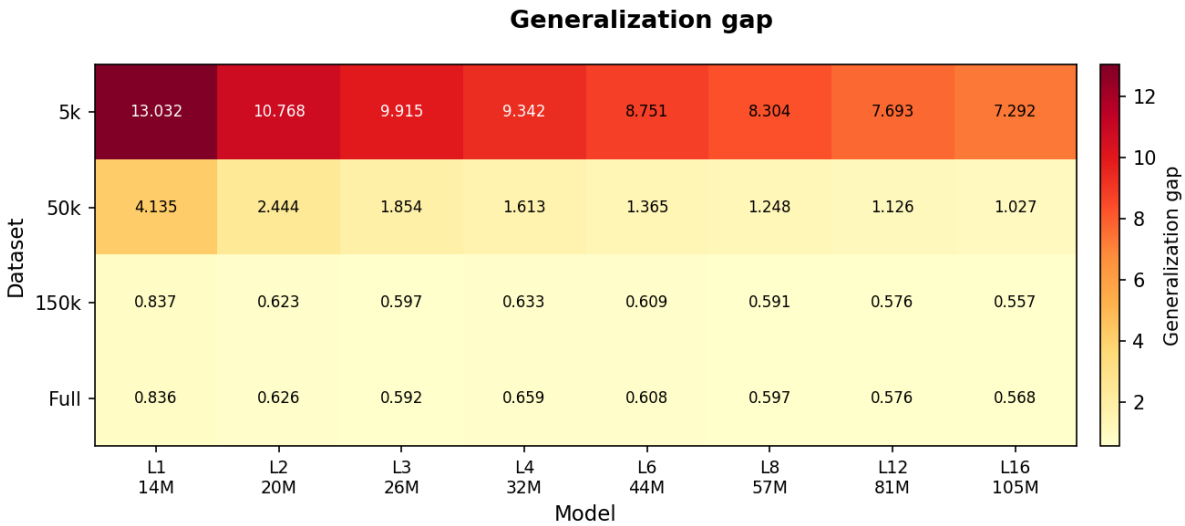

Figure 16: Generalization gap at step 100 k across all subset sizes and model capacities. For small subsets all models are in the overparameterization regime: the gap remains elevated but decreases monotonically with model size. For large subsets, small models have a large gap from underfitting; larger models show a decreasing gap as they approach interpolation.

Figures 15–16 summarize both regimes. Small subsets place all models on the overparameterization branch: every model overfits and generalization improves with capacity. Large subsets span the transition from underfitting to near-interpolation. Reproducing the double-descent peak requires a dataset that places some models in underfitting and larger ones in clear overfitting; the current capacity range does not achieve this within 100 k steps.

6 Conclusions

Input normalization mode does not affect Conformer-CTC performance or overfitting dynamics on LibriSpeech; any consistent CMVN mode is valid for Stage A and beyond.

The full-data capacity sweep shows monotonically decreasing test WER with model depth and no

double-descent peak within 100 k steps under clean labels. The val WER curve for $L \geq 12$ has not turned by the end of training, so these models are still in the descending phase of overparameterization.

The subset ablation reveals two complementary regimes. On small corpora (5k and 50k), every model interpolates the training set but larger models generalize better, confirming the overparameterization effect. On large corpora (150k and full), small models underfit and do not reach the interpolation threshold. No double-descent peak appears in either condition: small subsets give access only to the descending branch, while large subsets do not push the model family into clear overparameterization within 100 k steps.

Three directions remain for completing the double-descent characterization:

1. **Intermediate subset size.** A training set of approximately 75 k–100 k utterances could place the interpolation threshold between $L = 4$ and $L = 8$, making both branches of the curve accessible in a single sweep.
2. **Stage B: label noise.** Static label noise at 5%, 10%, and 20% corruption should raise the effective interpolation threshold for a given model size, potentially revealing the double-descent peak within the current model family on the full training set.
3. **Larger models or longer training.** Models beyond $L = 16$, or runs beyond 100 k steps, would push the largest models into clear overfitting and make the overparameterization regime unambiguous.

References

[1] Mikhail Belkin, Daniel Hsu, Siyuan Ma, and Soumik Mandal. Reconciling modern machine-learning practice and the classical bias–variance trade-off. *Proceedings of the National Academy of Sciences*, 116(32):15849–15854, July 2019. ISSN 1091-6490. doi: 10.1073/pnas.1903070116. URL `http://dx.doi.org/10.1073/pnas.1903070116`.

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